



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Class Test Examination, Spring-2024
Course Code: MATH-1107, Course Title: Mathematics-I
Time: 20 min Marks: 10

Section 1CF

(Answer any two (02) questions. Figures in the right margin indicates full marks)

- | | Marks | CLO | DL |
|--|-------|------|----|
| 1. Write down the following definitions | 5 | CLO2 | C2 |
| <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p>a) Partial Derivative</p> <p>c) Euler's Theorem</p> <p>e) Gamma Function</p> <p>g) Conditions of Extreme Values</p> <p>i) $\int_0^a f(x) dx = ?$</p> </div> <div style="width: 45%;"> <p>b) Homogeneous Function</p> <p>d) Maxima and Minima</p> <p>f) Beta Function</p> <p>h) $\int uv dx = ?$</p> <p>j) $\int_0^{\pi/2} \sin^m x \cos^n x dx = ?$</p> </div> </div> | | | |
| 2. Evaluate the integral $\int \frac{dx}{(2x+1)\sqrt{4x+3}}$ | 5 | CLO2 | C2 |
| 3. Find the arc length of the curve $y = \frac{2}{3}x^{\frac{3}{2}}$ from $x = 3$ to $x = 8$ about the x-axis. | 5 | CLO3 | C3 |



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